

Linear Relations

First Differences · Direct & Partial Variation · Slope · Intercepts · Equations of Lines · Standard Form

MTH1W · Class — Guided Practice

Name: _____

Date: _____

Goal: Represent, analyse and write linear relations using first differences, variation, slope, intercepts and equations.

Warm-up (quick recall)

1. Find the first differences for the y-values 5, 8, 11.

2. Is $y = 6x$ direct or partial variation?

3. Find the slope of the line through (0, 0) and (1, 4).

Examples & Practice

For each skill, study the worked **Example**, then solve the questions that follow. Difficulty rises within each set (L1 → L3). Answers are on the last page.

Patterns, Tables & First Differences

Example:

A table gives y-values 3, 7, 11, 15 for $x = 0, 1, 2, 3$. Are they linear?

First differences: $7-3 = 4$, $11-7 = 4$, $15-11 = 4$.

The first differences are constant (4), so the relation is linear.

4. The y-values in a table are 2, 5, 8, 11. Find the first differences and state whether the relation is linear. **[L1]**

5. A pattern gives the values 4, 9, 14, 19. Are the first differences constant? **[L1]**
6. For the y-values 1, 4, 9, 16, find the first differences and decide whether the relation is linear. **[L2]**
7. A linear relation has first differences all equal to 6, and $y = -2$ when $x = 0$. Write the y-values for $x = 0, 1, 2, 3$. **[L3]**

Direct vs Partial Variation

Example:

Classify $y = 5x$ and $y = 5x + 2$.

$y = 5x$ passes through $(0, 0)$, so it is direct variation.

$y = 5x + 2$ has y-intercept 2 (not 0), so it is partial variation.

8. State whether $y = 3x$ is direct or partial variation. **[L1]**

9. State whether $y = 2x + 7$ is direct or partial variation. **[L1]**
10. A taxi charges a \$3 flag fee plus \$2 per kilometre: $C = 2d + 3$. Is this direct or partial variation, and what is the initial value? **[L2]**
11. A partial variation has initial value 8 and rate of change 3. Write its equation and find y when $x = 5$. **[L3]**

Slope (Rise over Run, from Two Points)

Example:

Find the slope of the line through (1, 2) and (4, 11).

$$\text{slope} = (11 - 2) / (4 - 1) = 9 / 3.$$

$$\text{slope} = 3.$$

12. Find the slope of the line through (0, 0) and (2, 6). **[L1]**

13. Find the slope of the line through (1, 3) and (5, 11). **[L1]**

14. Find the slope of the line through (-2, 5) and (4, -7). **[L2]**

15. A line through (3, k) and (7, 20) has slope 3. Find k. **[L3]**

x- and y-Intercepts

Example:

Find the intercepts of $y = 2x - 6$.

y-intercept: let $x = 0 \rightarrow y = -6$, giving (0, -6).

x-intercept: let $y = 0 \rightarrow 0 = 2x - 6 \rightarrow x = 3$, giving (3, 0).

16. Find the y-intercept of $y = 4x + 5$. **[L1]**

17. Find the x-intercept of $y = x - 7$. **[L1]**
18. For $2x + 3y = 12$, find both the x-intercept and the y-intercept. **[L2]**
19. A line has x-intercept 4 and y-intercept 8. Find its slope and its equation. **[L3]**

Equation of a Line — Slope + a Point, and from Two Points

Example:

Find the equation of the line with slope 3 through (2, 5).

$y = 3x + b$, so $5 = 3(2) + b \rightarrow b = -1$.

$y = 3x - 1$.

20. Write the equation of the line with slope 2 and y-intercept 4. **[L1]**

21. Find the equation of the line with slope 4 that passes through (1, 7). **[L2]**
22. Find the equation of the line with slope -1 that passes through (0, -3). **[L2]**
23. Find the equation of the line through (2, 3) and (6, 15). **[L3]**

Standard Form $Ax + By + C = 0$, and Correlation & Linearity

Example:

Write $y = 2x + 5$ in standard form, and describe a rising scatter.

Move all terms to one side: $2x - y + 5 = 0$.

A scatter whose points rise to the right in a tight band shows a strong positive linear correlation.

24. Write $y = 3x - 4$ in the standard form $Ax + By + C = 0$. **[L1]**
25. Write $y = (1/2)x + 3$ in standard form with integer coefficients. **[L2]**
26. A scatter plot's points rise to the right and lie in a narrow band. Describe the correlation in words. **[L2]**
27. A scatter plot of daily temperature versus hot-chocolate sales shows points falling steeply to the right in a tight band. Describe the correlation and what it means. **[L3]**

Answers

Warm-up: W1) 3, 3 W2) Direct W3) 4

4. (L1) Differences 3, 3, 3 are constant, so the relation is linear.
5. (L1) Yes: 5, 5, 5 — constant, so it is linear.
6. (L2) Differences 3, 5, 7 are not constant, so the relation is non-linear.
7. (L3) Add 6 each step: -2, 4, 10, 16.
8. (L1) Direct variation (passes through the origin).
9. (L1) Partial variation (y-intercept $7 \neq 0$).
10. (L2) Partial variation; the initial value (fixed fee) is 3.
11. (L3) $y = 3x + 8$; at $x = 5$, $y = 3(5) + 8 = 23$.
12. (L1) slope = $6/2 = 3$.
13. (L1) slope = $(11-3)/(5-1) = 8/4 = 2$.
14. (L2) slope = $(-7-5)/(4-(-2)) = -12/6 = -2$.
15. (L3) $(20 - k)/(7 - 3) = 3 \rightarrow 20 - k = 12 \rightarrow k = 8$.
16. (L1) y-intercept = 5, at (0, 5).
17. (L1) $0 = x - 7 \rightarrow x = 7$, at (7, 0).
18. (L2) x-int: $y=0 \rightarrow 2x=12 \rightarrow (6, 0)$; y-int: $x=0 \rightarrow 3y=12 \rightarrow (0, 4)$.
19. (L3) Through (4, 0) and (0, 8): slope = $(8-0)/(0-4) = -2$; $y = -2x + 8$.
20. (L1) $y = 2x + 4$.
21. (L2) $7 = 4(1) + b \rightarrow b = 3$; $y = 4x + 3$.
22. (L2) y-intercept is -3; $y = -x - 3$.
23. (L3) slope = $(15-3)/(6-2) = 12/4 = 3$; $3 = 3(2)+b \rightarrow b = -3$; $y = 3x - 3$.
24. (L1) $3x - y - 4 = 0$.
25. (L2) Multiply by 2: $x - 2y + 6 = 0$.
26. (L2) A strong positive linear correlation.
27. (L3) A strong negative linear correlation: as temperature rises, hot-chocolate sales tend to fall.